

31.3-3

Prove Theorem 31.14.

31.3-4

Show that if p is prime and e is a positive integer, then

$$\phi(p^e) = p^{e-1}(p - 1).$$

31.3-5

Show that for any integer $n > 1$ and for any $a \in \mathbb{Z}_n^*$, the function $f_a : \mathbb{Z}_n^* \rightarrow \mathbb{Z}_n^*$ defined by $f_a(x) = ax \bmod n$ is a permutation of $\mathbb{Z}_n^*$.

31.4 Solving modular linear equations

We now consider the problem of finding solutions to the equation

$$ax = b \pmod{n}, \tag{31.26}$$

where $a > 0$ and $n > 0$. This problem has several applications. For example, we'll use it in Section 31.7 as part of the procedure to find keys in the RSA public-key cryptosystem. We assume that a , b , and n are given, and we wish to find all values of x , modulo n , that satisfy equation (31.26). The equation may have zero, one, or more than one such solution.

Let $\langle a \rangle$ denote the subgroup of $\mathbb{Z}_n$ generated by a . Since $\langle a \rangle = \{a^{(x)} : x > 0\} = \{ax \bmod n : x > 0\}$, equation (31.26) has a solution if and only if $[b] \in \langle a \rangle$. Lagrange's theorem (Theorem 31.15) tells us that $|\langle a \rangle|$ must be a divisor of n . The following theorem gives us a precise characterization of $\langle a \rangle$.

Theorem 31.20

For any positive integers a and n , if $d = \gcd(a, n)$, then we have

$$\begin{aligned} \langle a \rangle &= \langle d \rangle \\ &= \{0, d, 2d, \dots, ((n/d) - 1)d\} \end{aligned}$$

in $\mathbb{Z}_n$, and thus

$$|\langle a \rangle| = n/d.$$

Proof We begin by showing that $d \in \langle a \rangle$. Recall that EXTENDED-EUCLID(a, n) returns a triple (d, x, y) such that $ax + ny = d$. Thus, $ax = d \pmod{n}$, so that $d \in \langle a \rangle$. In other words, d is a multiple of a in $\mathbb{Z}_n$.

Since $d \in \langle a \rangle$, it follows that every multiple of d belongs to $\langle a \rangle$, because any multiple of a multiple of a is itself a multiple of a . Thus, $\langle a \rangle$ contains every element in $\{0, d, 2d, \dots, ((n/d) - 1)d\}$. That is, $\langle d \rangle \subseteq \langle a \rangle$.

We now show that $\langle a \rangle \subseteq \langle d \rangle$. If $m \in \langle a \rangle$, then $m = ax \pmod{n}$ for some integer x , and so $m = ax + ny$ for some integer y . Because $d = \gcd(a, n)$, we know that $d \mid a$ and $d \mid n$, and so $d \mid m$ by equation (31.4). Therefore, $m \in \langle d \rangle$.

Combining these results, we have that $\langle a \rangle = \langle d \rangle$. To see that $|\langle a \rangle| = n/d$, observe that there are exactly n/d multiples of d between 0 and $n - 1$, inclusive. ■

Corollary 31.21

The equation $ax = b \pmod{n}$ is solvable for the unknown x if and only if $d \mid b$, where $d = \gcd(a, n)$.

Proof The equation $ax = b \pmod{n}$ is solvable if and only if $[b] \in \langle a \rangle$, which is the same as saying

$$(b \pmod{n}) \in \{0, d, 2d, \dots, ((n/d) - 1)d\},$$

by Theorem 31.20. If $0 \leq b < n$, then $b \in \langle a \rangle$ if and only if $d \mid b$, since the members of $\langle a \rangle$ are precisely the multiples of d . If $b < 0$ or $b \geq n$, the corollary then follows from the observation that $d \mid b$ if and only if $d \mid (b \pmod{n})$, since b and $b \pmod{n}$ differ by a multiple of n , which is itself a multiple of d . ■

Corollary 31.22

The equation $ax = b \pmod{n}$ either has d distinct solutions modulo n , where $d = \gcd(a, n)$, or it has no solutions.

Proof If $ax = b \pmod{n}$ has a solution, then $b \in \langle a \rangle$. By Theorem 31.17, $\text{ord}(a) = |\langle a \rangle|$, and so Corollary 31.18 and Theorem 31.20 imply that the sequence $ai \pmod{n}$, for $i = 0, 1, \dots$, is periodic with period $|\langle a \rangle| = n/d$. If $b \in \langle a \rangle$, then b appears exactly d times in the sequence $ai \pmod{n}$, for $i = 0, 1, \dots, n-1$, since the length- (n/d) block of values $\langle a \rangle$ repeats exactly d times as i increases from 0 to $n-1$. The indices x of the d positions for which $ax \pmod{n} = b$ are the solutions of the equation $ax = b \pmod{n}$. ■

Theorem 31.23

Let $d = \gcd(a, n)$, and suppose that $d = ax' + ny'$ for some integers x' and y' (for example, as computed by EXTENDED-EUCLID). If $d \mid b$, then the equation $ax = b \pmod{n}$ has as one of its solutions the value x_0 , where

$$x_0 = x'(b/d) \pmod{n}.$$

Proof We have

$$\begin{aligned} ax_0 &= ax'(b/d) \pmod{n} \\ &= d(b/d) \pmod{n} \quad (\text{because } ax' = d \pmod{n}) \\ &= b \pmod{n}, \end{aligned}$$

and thus x_0 is a solution to $ax = b \pmod{n}$. ■

Theorem 31.24

Suppose that the equation $ax = b \pmod{n}$ is solvable (that is, $d \mid b$, where $d = \gcd(a, n)$) and that x_0 is any solution to this equation. Then, this equation has exactly d distinct solutions, modulo n , given by $x_i = x_0 + i(n/d)$ for $i = 0, 1, \dots, d-1$.

Proof Because $n/d > 0$ and $0 \leq i(n/d) < n$ for $i = 0, 1, \dots, d-1$, the values $x_0, x_1, \dots, x_{d-1}$ are all distinct, modulo n . Since x_0 is a solution

of $ax = b \pmod{n}$, we have $ax_0 \pmod{n} = b \pmod{n}$. Thus, for $i = 0, 1, \dots, d-1$, we have

$$\begin{aligned} ax_i \pmod{n} &= a(x_0 + in/d) \pmod{n} \\ &= (ax_0 + ain/d) \pmod{n} \\ &= ax_0 \pmod{n} \text{ (because } d \mid a \text{ implies that } ain/d \text{ is a multiple of } n) \\ &= b \pmod{n}, \end{aligned}$$

and hence $ax_i = b \pmod{n}$, making x_i a solution, too. By Corollary 31.22, the equation $ax = b \pmod{n}$ has exactly d solutions, so that $x_0, x_1, \dots, x_{d-1}$ must be all of them. ■

We have now developed the mathematics needed to solve the equation $ax = b \pmod{n}$. The procedure MODULAR-LINEAR-EQUATION-SOLVER prints all solutions to this equation. The inputs a and n are arbitrary positive integers, and b is an arbitrary integer.

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MODULAR-LINEAR-EQUATION-SOLVER( $a, b, n$ )
1 ( $d, x', y'$ ) = EXTENDED-EUCLID( $a, n$ )
2 if  $d \mid b$ 
3    $x_0 = x'(b/d) \pmod{n}$ 
4   for  $i = 0$  to  $d - 1$ 
5     print  $(x_0 + i(n/d)) \pmod{n}$ 
6 else print “no solutions”
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As an example of the operation of MODULAR-LINEAR-EQUATION-SOLVER, consider the equation $14x = 30 \pmod{100}$ (and thus $a = 14$, $b = 30$, and $n = 100$). Calling EXTENDED-EUCLID in line 1 gives $(d, x', y') = (2, -7, 1)$. Since $2 \mid 30$, lines 3–5 execute. Line 3

computes $x_0 = (-7)(15) \bmod 100 = 95$. The **for** loop of lines 4–5 prints the two solutions, 95 and 45.

The procedure MODULAR-LINEAR-EQUATION-SOLVER works as follows. The call to EXTENDED-EUCLID in line 1 returns a triple (d, x', y') such that $d = \gcd(a, n)$ and $d = ax' + ny'$. Therefore, x' is a solution to the equation $ax' = d \pmod{n}$. If d does not divide b , then the equation $ax = b \pmod{n}$ has no solution, by Corollary 31.21. Line 2 checks to see whether $d \mid b$, and if not, line 6 reports that there are no solutions. Otherwise, line 3 computes a solution x_0 to $ax = b \pmod{n}$, as Theorem 31.23 suggests. Given one solution, Theorem 31.24 states that adding multiples of (n/d) , modulo n , yields the other $d - 1$ solutions. The **for** loop of lines 4–5 prints out all d solutions, beginning with x_0 and spaced n/d apart, modulo n .

MODULAR-LINEAR-EQUATION-SOLVER performs $O(\lg n + \gcd(a, n))$ arithmetic operations, since EXTENDED-EUCLID performs $O(\lg n)$ arithmetic operations, and each iteration of the **for** loop of lines 4–5 performs a constant number of arithmetic operations.

The following corollaries of Theorem 31.24 give specializations of particular interest.

Corollary 31.25

For any $n > 1$, if $\gcd(a, n) = 1$, then the equation $ax = b \pmod{n}$ has a unique solution, modulo n .

■

If $b = 1$, a common case of considerable interest, the x that solves the equation is a *multiplicative inverse* of a , modulo n .

Corollary 31.26

For any $n > 1$, if $\gcd(a, n) = 1$, then the equation $ax = 1 \pmod{n}$ has a unique solution, modulo n . Otherwise, it has no solution.

■

Thanks to Corollary 31.26, the notation $a^{-1} \bmod n$ refers to *the* multiplicative inverse of a , modulo n , when a and n are relatively prime.

If $\gcd(a, n) = 1$, then the unique solution to the equation $ax = 1 \pmod{n}$ is the integer x returned by EXTENDED-EUCLID, since the equation

$$\gcd(a, n) = 1 = ax + ny$$

implies $ax = 1 \pmod{n}$. Thus, EXTENDED-EUCLID can compute $a^{-1} \pmod{n}$ efficiently.

Exercises

31.4-1

Find all solutions to the equation $35x = 10 \pmod{50}$.

31.4-2

Prove that the equation $ax = ay \pmod{n}$ implies $x = y \pmod{n}$ whenever $\gcd(a, n) = 1$. Show that the condition $\gcd(a, n) = 1$ is necessary by supplying a counterexample with $\gcd(a, n) > 1$.

31.4-3

Consider the following change to line 3 of the procedure MODULAR-LINEAR-EQUATION-SOLVER:

$$3 \quad x_0 = x'(b/d) \pmod{n/d}$$

With this change, will the procedure still work? Explain why or why not.

★ 31.4-4

Let p be prime and $f(x) = (f_0 + f_1x + \cdots + f_tx^t) \pmod{p}$ be a polynomial of degree t , with coefficients f_i drawn from $\mathbb{Z}_p$. We say that $a \in \mathbb{Z}_p$ is a **zero** of f if $f(a) = 0 \pmod{p}$. Prove that if a is a zero of f , then $f(x) = (x - a)g(x) \pmod{p}$ for some polynomial $g(x)$ of degree $t - 1$. Prove by induction on t that if p is prime, then a polynomial $f(x)$ of degree t can have at most t distinct zeros modulo p .

31.5 The Chinese remainder theorem